IPL DATASET ANALYSIS USING PYTHON

PROJECT REPORT

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ABSTRACT

This project is about studying IPL cricket data using Python. The Indian Premier League (IPL) has a large amount of information about matches, teams, and players. By analyzing this data, we can understand which teams have won more matches, which players have scored more runs, and how teams have performed while batting.

In this project, I use Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn. Pandas is used to read and organize the data, while Matplotlib and Seaborn are used to create graphs. The project includes loading the dataset, checking missing values, removing duplicate records, calculating summaries, and presenting the results using charts.

This project helped me understand how Python can be used to convert raw sports data into useful information.

1. INTRODUCTION

Data science is used to find useful information from data. Sports data is one example because every match contains information about teams, players, runs, wickets, and results. The Indian Premier League is a popular cricket tournament, and its data can be analyzed to understand team and player performance.

For this project, I selected IPL Dataset Analysis using Python. My main aim is to study team performance, player performance, and top run scorers. I have used a dataset containing match details and ball-by-ball records. The analysis is performed using basic Python code, tables, and graphs so that the results are easy to understand.

2. OBJECTIVES

The main objectives of this project are:

1. To load IPL match data into Python.
2. To check the dataset for missing values and duplicate records.
3. To find the teams with the highest number of wins.
4. To identify the top run scorers in the dataset.
5. To compare total runs and average innings scores of teams.
6. To display the results using graphs.
7. To understand how data analysis can help explain sports performance.

3. SOFTWARE AND TOOLS USED

The following tools and libraries are used in this project:

- Python: Used for writing and executing the program.
- Jupyter Notebook: Used to write code and display results.
- Pandas: Used to load, clean, and analyze the dataset.
- NumPy: Used for numerical calculations.
- Matplotlib: Used to create graphs and charts.
- Seaborn: Used to create clear and attractive visualizations.

4. DATASET DESCRIPTION

The project uses two CSV files: `matches.csv` and `deliveries.csv`.

The `matches.csv` file contains information about each match, such as the season, participating teams, match winner, toss winner, and Player of the Match.

The `deliveries.csv` file contains ball-by-ball information, including batter names, batting teams, runs scored, and total runs on each delivery.

These datasets help analyze both team performance and individual player performance.

Dataset source: IPL Complete Dataset (2008–2020), Kaggle.

https://www.kaggle.com/datasets/patrickb1912/ipl-complete-dataset-20082020

5. METHODOLOGY

I followed the steps below to complete the project:

1. Downloaded the IPL dataset.
2. Imported the required Python libraries.
3. Loaded the CSV files using Pandas.
4. Checked the dataset structure, missing values, and duplicate records.
5. Calculated team wins and total runs scored by each team.
6. Calculated the total runs scored by each batter.
7. Identified the top ten run scorers.
8. Created graphs to compare team and player performance.
9. Studied the results and wrote the observations.

6. ANALYSIS PERFORMED

6.1 Team Performance Analysis

Team performance is analyzed by counting the number of matches won by each team. The team with more wins has achieved more victories in the selected dataset.

I also calculated the total runs scored by each batting team. This helps compare the overall batting performance of different teams. Average innings scores are also useful because teams may have played different numbers of matches.

A bar chart is used to display the total wins of each team and make comparisons easier.

6.2 Top Run Scorers Analysis

In this section, I analyzed individual player performance based on total runs scored.

The delivery records are grouped by batter, and the runs scored by each batter are added together. The players are then sorted in descending order of their total runs.

The top ten run scorers are displayed using a bar chart. This helps identify the players who have scored the most runs in the selected dataset.

However, total runs alone do not completely measure a player's performance. Strike rate, balls faced, and innings played should also be considered.

6.3 Player of the Match Analysis

The Player of the Match award is another way to study individual performance.

I counted how many times each player received this award and displayed the top players using a chart. This provides an additional view of players who have been recognized for their performances in matches.

However, this award is only one measure of performance and does not represent every contribution made by a player.

6.4 Toss and Match Result Analysis

In this section, I compared the toss-winning team with the match-winning team.

The percentage of matches in which the toss winner also won the match is calculated using the available records. This helps us understand the relationship between toss results and match outcomes.

Winning the toss does not guarantee winning the match because the final result depends on many factors, including player performance, pitch conditions, and match strategy.

7. RESULTS AND OBSERVATIONS

After executing the Python notebook, the following results are generated:

1. Number of matches played in each season.
2. Total matches won by each team.
3. Top ten run scorers.
4. Total runs scored by each batting team.
5. Average innings score of each team.
6. Players with the highest number of Player of the Match awards.
7. Toss decisions and the percentage of matches won by toss winners.

The graphs make it easier to understand the differences between teams and players.

The exact rankings and numerical values should be filled in after running the notebook with the selected IPL dataset.

8. ANALYSIS AND INFERENCE

From this project, the following observations can be made after reviewing the generated results:

- The team-win chart helps identify the teams with the highest number of victories.
- Total runs show the overall batting output of each team.
- Average innings scores help compare team scoring performance more fairly.
- The top run-scorer chart identifies players with the highest accumulated runs.
- Player of the Match awards provide additional information about individual match performance.
- Toss results can be compared with match outcomes, but they do not prove that winning the toss causes a team to win.
- The findings depend on the seasons and records included in the dataset.

This analysis shows how basic data science techniques can be used to understand sports statistics.

9. CONCLUSION

Through this project, I learned how to use Python for basic data analysis. I practiced loading CSV files, checking data quality, grouping records, calculating totals and averages, and creating graphs.

The project helped me understand how IPL data can be used to compare teams and players. I also learned that visualizations make it easier to understand data and that conclusions should be based on actual results.

As a beginner in data science, this project gave me practical experience with Python libraries and the steps involved in analyzing a real-world dataset.

10. LIMITATIONS AND FUTURE SCOPE

The project mainly focuses on basic statistics and visualizations. The results depend on the quality and time period of the dataset. Some records may contain missing values, and team names may vary across different seasons.

In the future, the project can be improved by:

- Adding detailed bowling performance analysis.
- Comparing player performance across individual seasons.
- Calculating batting strike rates and bowling economy rates.
- Predicting match outcomes using machine learning.